ML Homework 3 Guide

1. Save a copy of this ipynb file in your GoogleDrive or PC.
2. Edit the name of this code from "HW3.ipynb" to "HW3_(your name).ipynb"
3. Fill out the code cells below according to descriptions.
4. Save and upload to BrightSpace. (DO NOT clear the outputs of your code)
5. Convert the .ipynb file to PDF file and upload together

[HW 3] Customize the ANN

1. Load dataset resulted from ML3_Code2 (FeatureSelected) and create corresponding labels referring to ML5_Code1.
2. Design and train a customized ANN model according to the following instructions:

- Set the learning rate to 0.01.
- Set the number of hidden layers to 3.
- Set the number of neurons in the first hidden layer to 15.
- Set the number of neurons in the second hidden layer to 10.
- Set the number of neurons in the third hidden layer to 5.
- Set the number of training epochs to 300.
- Set the activation function differently from the ML5 code by referring to the link provided in the code. (Few functions may not work due to compatibility issues, please try other functions in that case.)
- Set the loss function and optimization algorithm differently from the ML5 code by referring to the link provided. (Few functions may not work due to compatibility issues, please try other functions in that case.)

3. Plot a graph of the training process (Loss, Accuracy) at the end.

```
# Start with importing packages
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
# Import 'Tensorflow' pakage
import tensorflow as tf
from tensorflow import keras
```

```
# Check if a GPU(in Google server) is allocated
device_name = tf.test.gpu_device_name()
print('Found GPU at: {}'.format(device_name))
```

```
Found GPU at:
```

```
# Acess to google drive
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
FeatureSelected = pd.read_csv('/content/drive/MyDrive/Colab Notebooks/HW/FeatureSelected.csv', header
FeatureSelected = FeatureSelected.T
FeatureSelected.shape
```

```
(360, 30)
```

```
# Standardize feature values
from sklearn.preprocessing import StandardScaler
FeatureSelected_std = StandardScaler().fit_transform(FeatureSelected)

# Split Normal and Abnormal sets
NoOfData = int(FeatureSelected_std.shape[0]/2)

NormalSet = FeatureSelected_std[:NoOfData,:]
```

```
AbnormalSet = FeatureSelected_std[NoOfData:,:]
NormalSet.shape, AbnormalSet.shape
```

```
((180, 30), (180, 30))
```

```
# Train/Test split
from sklearn.model_selection import train_test_split

TestData_Ratio = 0.2

TrainData_Nor, TestData_Nor = train_test_split(NormalSet, test_size=TestData_Ratio, random_state=77)
TrainData_Abn, TestData_Abn = train_test_split(AbnormalSet, test_size=TestData_Ratio, random_state=7
```

```
# One-hot labels
TrainLabel_Nor = np.zeros((TrainData_Nor.shape[0],2))
TrainLabel_Abn = np.ones((TrainData_Abn.shape[0],2))
TestLabel_Nor = np.zeros((TestData_Nor.shape[0],2))
TestLabel_Abn = np.ones((TestData_Abn.shape[0],2))

TrainLabel_Nor[:,0] = 1
TrainLabel_Abn[:,0] = 0
TestLabel_Nor[:,0] = 1
TestLabel_Abn[:,0] = 0
```

```
# Combine data
TrainData = np.concatenate([TrainData_Nor,TrainData_Abn],axis=0)
TestData = np.concatenate([TestData_Nor,TestData_Abn],axis=0)

TrainLabel = np.concatenate([TrainLabel_Nor,TrainLabel_Abn],axis=0)
TestLabel = np.concatenate([TestLabel_Nor,TestLabel_Abn],axis=0)
```

```
# Hyperparameters
learningRate = 0.01
Epoch = 300
H1=15
H2=10
H3=5
# Build customized ANN model
def ANN_model(input_data):

    keras.backend.clear_session()
    model = keras.Sequential()

    model.add(keras.layers.InputLayer(input_shape=(input_data.shape[1],)))
    # Hidden Layer 1
    model.add(keras.layers.Dense(H1, activation='tanh'))
    # Hidden Layer 2
    model.add(keras.layers.Dense(H2, activation='tanh'))
    # Hidden Layer 3
    model.add(keras.layers.Dense(H3, activation='tanh'))
    # Output Layer
    model.add(keras.layers.Dense(2, activation='softmax'))

    # Compile model
    model.compile(
        optimizer = keras.optimizers.SGD(learning_rate=learningRate),
        loss = keras.losses.MeanSquaredError(),
        metrics = ['accuracy'])

    return model
```

```
# Create model
AnnModel = ANN_model(TrainData)
# Check architecture
AnnModel.summary()
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/input_layer.py:27: UserWarning: Argument
  warnings.warn(
```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 15)	465
dense_1 (Dense)	(None, 10)	160
dense_2 (Dense)	(None, 5)	55
dense_3 (Dense)	(None, 2)	12

Total params: 692 (2.70 KB)
Trainable params: 692 (2.70 KB)
Non-trainable params: 0 (0.00 B)

```
# Train model
tf.random.set_seed(777)
TrainingHistory = AnnModel.fit(TrainData,TrainLabel,epochs = Epoch,verbose = 1)
```

```
Epoch 1/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 1s 7ms/step - accuracy: 0.4489 - loss: 0.2918
Epoch 2/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.8890 - loss: 0.1401
Epoch 3/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9769 - loss: 0.0822
Epoch 4/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.9861 - loss: 0.0571
Epoch 5/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.9861 - loss: 0.0436
Epoch 6/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.9908 - loss: 0.0353
Epoch 7/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0296
Epoch 8/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0255
Epoch 9/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 1.0000 - loss: 0.0224
Epoch 10/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0200
Epoch 11/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0180
Epoch 12/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0164
Epoch 13/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0151
Epoch 14/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0139
Epoch 15/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0130
Epoch 16/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0121
Epoch 17/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0114
Epoch 18/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0107
Epoch 19/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0101
Epoch 20/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0096
Epoch 21/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0091
Epoch 22/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0087
Epoch 23/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0083
Epoch 24/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0080
Epoch 25/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 1.0000 - loss: 0.0076
Epoch 26/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 1.0000 - loss: 0.0073
Epoch 27/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0071
Epoch 28/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0068
Epoch 29/300
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 1.0000 - loss: 0.0066
```

```
#Check the training process (Loss, Accuracy)
Loss, Accuracy = AnnModel.evaluate(TestData, TestLabel, verbose=0)
print("Test Loss:", Loss)
print("Test Accuracy:", Accuracy)
```

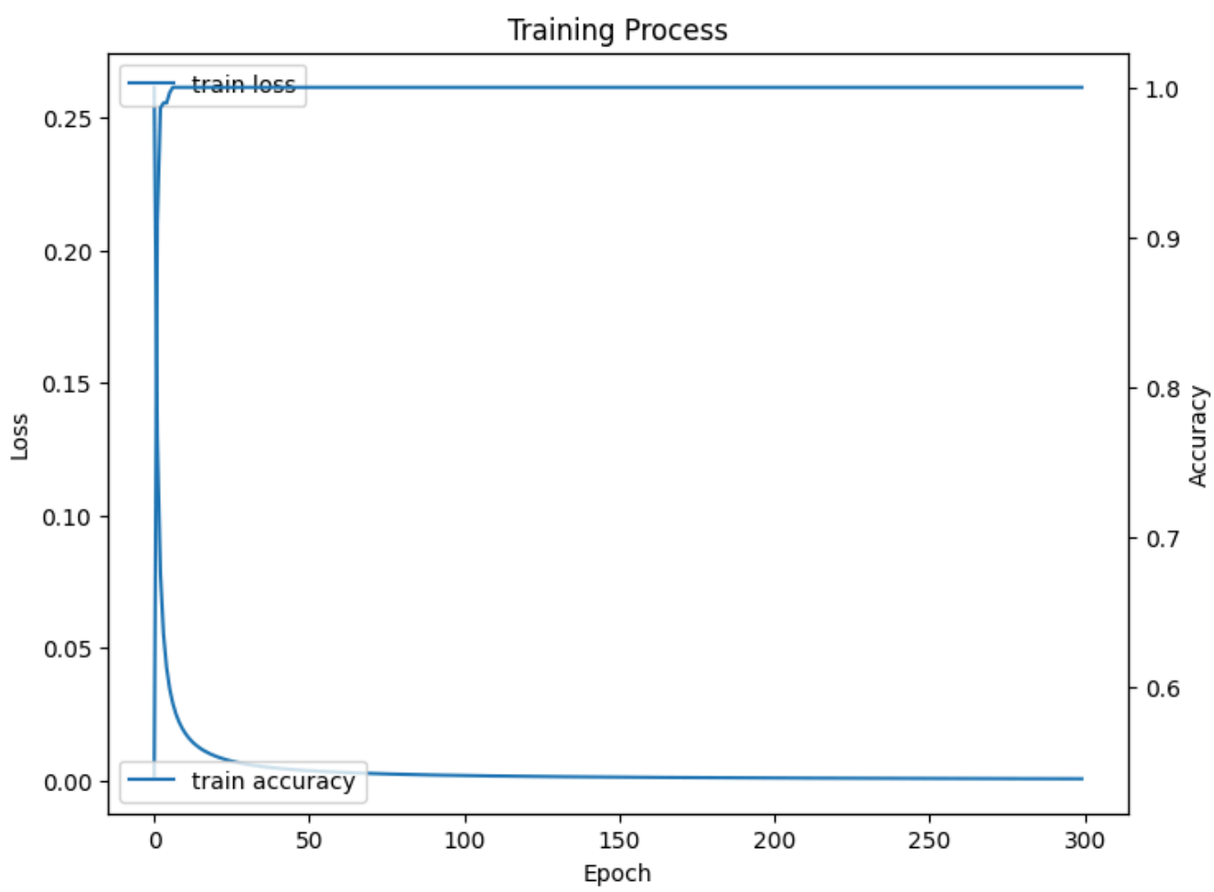
```
Test Loss: 0.005988976452499628
Test Accuracy: 0.9861111044883728
```

```
# Plot training process
fig, loss_ax = plt.subplots(figsize=(8,6))
acc_ax = loss_ax.twinx()

loss_ax.plot(TrainingHistory.history['loss'], label='train loss')
loss_ax.set_xlabel('Epoch')
loss_ax.set_ylabel('Loss')
loss_ax.legend(loc='upper left')

acc_ax.plot(TrainingHistory.history['accuracy'], label='train accuracy')
acc_ax.set_ylabel('Accuracy')
acc_ax.legend(loc='lower left')

plt.title("Training Process")
plt.show()
```



ML5 and ML6 Summary and Deliverables

Answer the following questions for your achievements

Q1. Please summarize ML5 and ML6.

ML5 Summary:

ML5 classifies machine conditions using selected statistical features and an ANN. The top 30 features are standardized, split into training (80%) and testing (20%) sets, and one-hot encoded ([1,0] = normal, [0,1] = abnormal). A Keras ANN with two ReLU hidden layers and a softmax output is trained for 200 epochs using Adam optimizer. Model performance is evaluated via accuracy and loss, then saved for future use.

ML6 Summary:

ML6 detects faults from raw acceleration signals using CNNs. Vibration data is converted to spectrograms via STFT, reshaped for CNN input, and one-hot encoded. The CNN includes convolution, max-pooling, flatten, and

dense layers, trained for 1000 epochs. Performance is evaluated using accuracy, precision, recall, F1 score, and confusion matrix. The trained model is saved for predictions.

Q2. What skills did you have to develop to accomplish this project?

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1. Data preprocessing and feature selection
2. Neural network and CNN model design and training
3. Performance evaluation and model saving/loading

Q3. What aspects of this project were the most beneficial for your learning?

- Applying machine learning techniques to real sensor data
- Understanding ANN and CNN architectures and training processes
- Converting raw signals into meaningful features for classification

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Q4. What challenges did you encounter in completing the project?

- Handling and preprocessing large, noisy datasets
- Designing and tuning ANN and CNN architectures for accurate classification
- Managing long training times and avoiding overfitting

Q5. How did you overcome the challenges or remedy the problems encountered?

- Applied data normalization, feature selection, and one-hot encoding to clean and organize data
- Experimented with different network architectures, activation functions, and hyperparameters
- Used proper train-test splits, early stopping, and evaluation metrics to ensure model reliability

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